Arista Access Node Routing Protocol Flap Resolution Procedure

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4278

Description

The Arista access node, situated at the Electronic City Hub, designated as ELEC-ACC-01, is susceptible to routing protocol flaps due to the intrinsic complexities of BGP, OSPF, and ISIS protocols. Specifically, BGP route churn occurs when the node experiences fluctuations in prefix advertisements, leading to transient routing table instabilities. Similarly, OSPF adjacency loss can arise from hello packet mismatches or dead interval expirations, while ISIS LSP flooding may be triggered by excessive link-state updates. These phenomena can be attributed to the inherent mechanics of distance-vector and link-state protocols, where routing information dissemination and convergence times play a critical role. The Arista EOS, with its robust implementation of these protocols, provides a comprehensive set of tools to diagnose and rectify such issues. The warning severity alarm, ALARM_4278, serves as an indicator of such protocol flaps, necessitating a prompt and methodical response to ensure network stability and prevent potential outages.

Pre-requisite Checks

- Verify redundant link status to ensure failover capabilities
- Dispatch remote hands to Electronic City Hub for on-site assistance
- Confirm Arista EOS software version and compatibility with diagnostic tools

Potential Root Causes

Cause	Probability
BGP neighbor configuration mismatch	High
OSPF network type mismatch	Medium
ISIS area or domain configuration errors	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp neighbor, Expected Output: BGP neighbor information, including state and prefix counts, Description: This command isolates BGP neighbor relationships, allowing for the identification of potential configuration discrepancies or neighbor state issues that may be contributing to the routing protocol flap.

Command: show ip ospf interface, Expected Output: OSPF interface information, including network type and hello intervals, Description: This command provides insight into OSPF interface configurations, facilitating the detection of network type mismatches or other configuration anomalies that may be causing OSPF adjacency loss.
```

```
Command: show isis database, Expected Output: ISIS link-state database information, including LSP IDs and sequence numbers, Description: This command enables the examination of the ISIS link-state database, allowing for the identification of potential issues with LSP flooding or other ISIS-related problems that may be contributing to the routing protocol flap.
```

Resolution Steps

Step 1: Step: BGP Neighbor Configuration Correction, Command: neighbor description , Description: Correct BGP neighbor configuration discrepancies by updating the neighbor description, which may help resolve BGP route churn issues by ensuring consistent neighbor configurations.

Step 2: Step: OSPF Network Type Adjustment, Command: ip ospf network , Description: Adjust the OSPF network type to match the configured network type on adjacent routers, which can help prevent OSPF adjacency loss by ensuring consistent network type configurations.

Step 3: Step: ISIS Area or Domain Configuration Update, Command: isis area , Description: Update the ISIS area or domain configuration to ensure consistency with adjacent routers, which can help prevent ISIS LSP flooding by ensuring that LSPs are properly propagated within the correct area or domain.

Post-Check Verification

```
Command: show ip bgp neighbor, Expected Output: Updated BGP neighbor information, reflecting the corrected configuration, Description: Verify that the BGP neighbor configuration has been successfully updated and that the neighbor state is stable.

Command: show ip ospf interface, Expected Output: Updated OSPF interface information, reflecting the adjusted network type, Description: Verify that the OSPF network type has been successfully adjusted and that OSPF adjacency is stable.

Command: show isis database, Expected Output: Updated ISIS link-state database information, reflecting the corrected area or domain configuration, Description: Verify that the ISIS area or domain configuration has been successfully updated and that LSP flooding has ceased.
```

Roll-back Procedures

- Step: Revert BGP Neighbor Configuration, Command: no neighbor description, Description: Revert the BGP neighbor configuration to its original state, in case the updated configuration introduced unforeseen issues.

- Step: Revert OSPF Network Type, Command: no ip ospf network, Description: Revert the OSPF network type to its original configuration, in case the adjusted network type introduced unforeseen issues.

- Step: Revert ISIS Area or Domain Configuration, Command: no isis area, Description: Revert the ISIS area or domain configuration to its original state, in case the updated configuration introduced unforeseen issues.